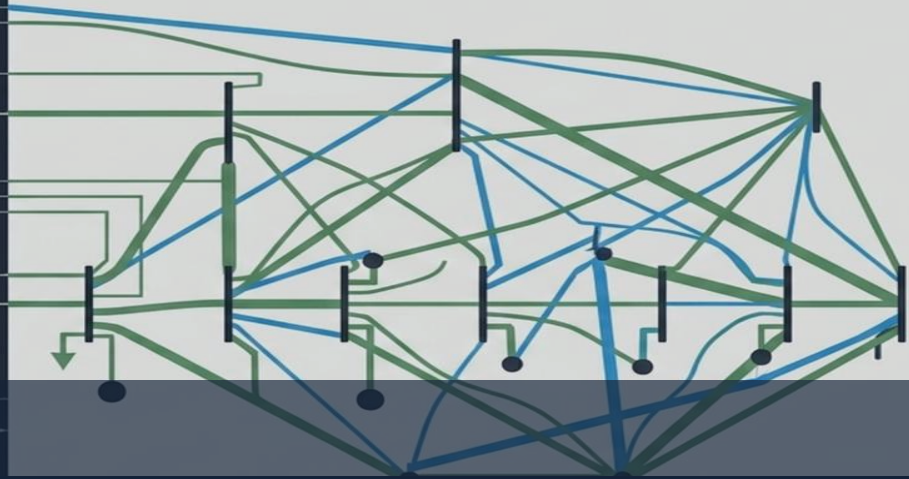
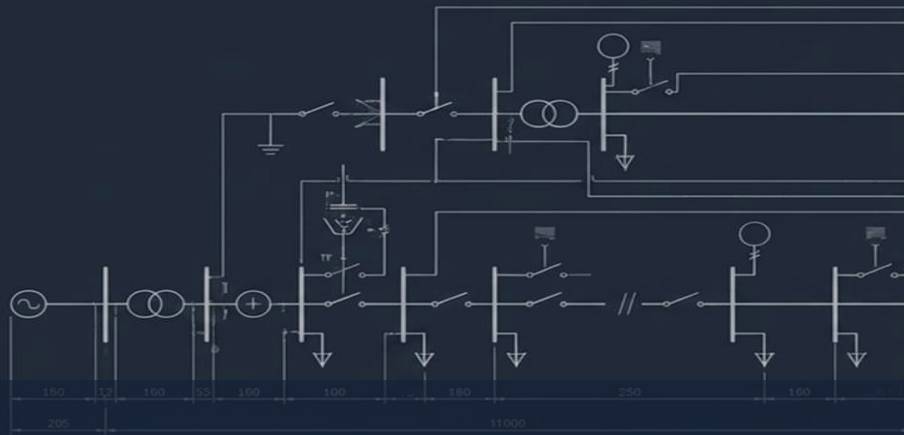




PIGNN-L5

Physics-Informed Graph Neural Networks for Power Flow in Renewable-Rich Active Distribution Networks

Noman Khan, Vlad-Alexandru Sanduleac, Shahariya R. R. Puthanpurayil, Remus Teodorescu (Fellow, IEEE)

AAU Energy, Aalborg University, Denmark · Prepared for IEEE Transactions on Smart Grid

MOTIVATION

DER proliferation breaks the legacy distribution-analysis toolchain

Rooftop PV and BESS turn passive feeders into active networks: voltage fluctuation, reverse power flow, and thermal-loading variation now dominate operating behaviour.

Renewable-rich operation generates massive populations of feasible-but-stressed operating points that must be screened repeatedly for monitoring, hosting-capacity assessment, Volt/VAR advisory, and contingency screening.

Newton Raphson (pandapower)

Accuracy reference (tol. 10^{-6} p.u.) but 62.3 ms per snapshot is too slow for real-time, edge-deployable stochastic screening.

2.88 M

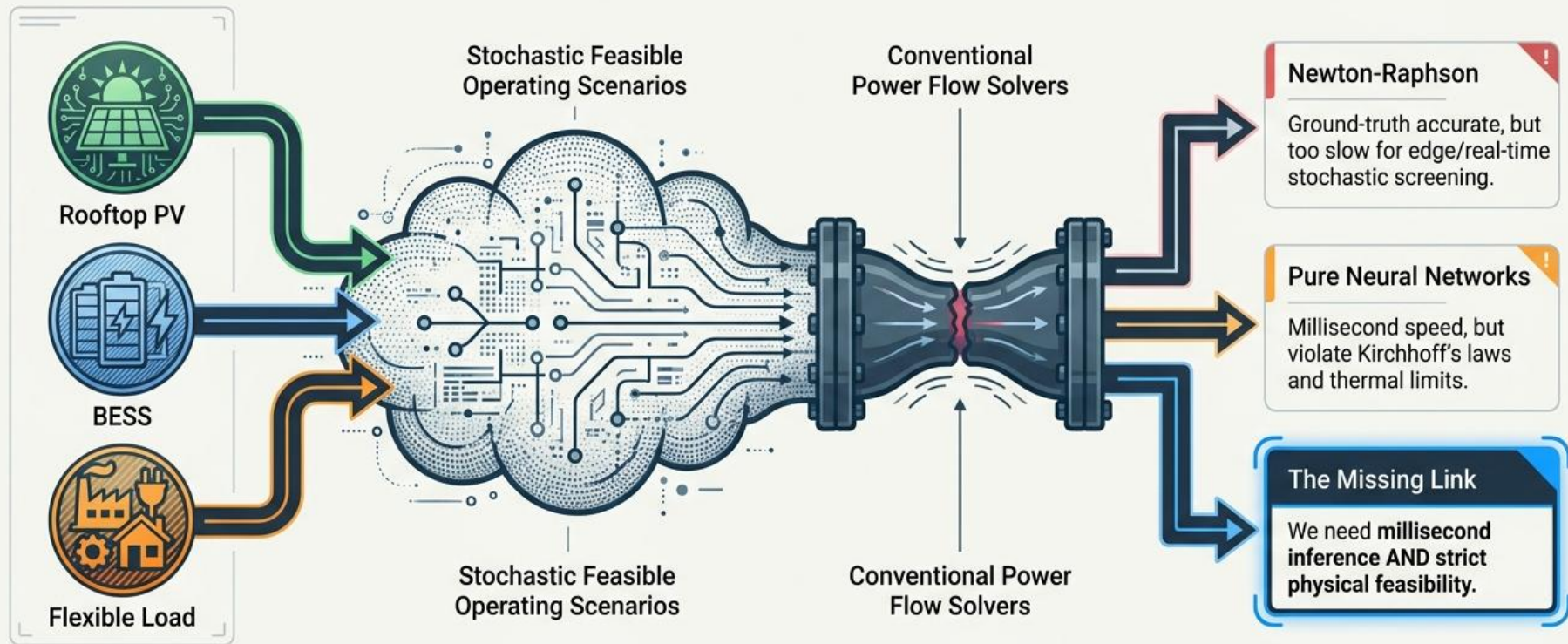
operating points needed for one 30,000-scenario stochastic study (96 timesteps / day)

The engineering question is not how to approximate AC power flow quickly but how to do so while preserving feeder topology and physical feasibility.

Pure neural surrogates

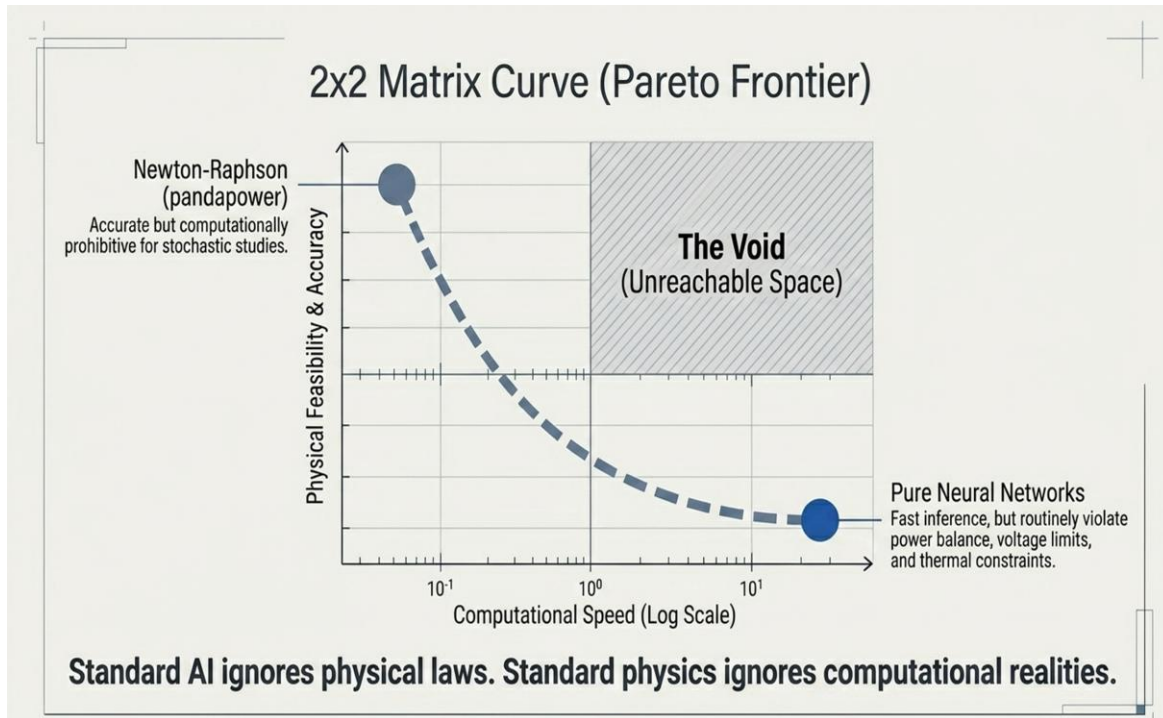
Millisecond inference but routinely violate power balance, voltage limits, slack consistency and thermal constraints.

The proliferation of DERs breaks the legacy distribution-analysis toolchain.



The central engineering question is not how to approximate AC power flow quickly, but how to do so while preserving feeder topology and physical feasibility.

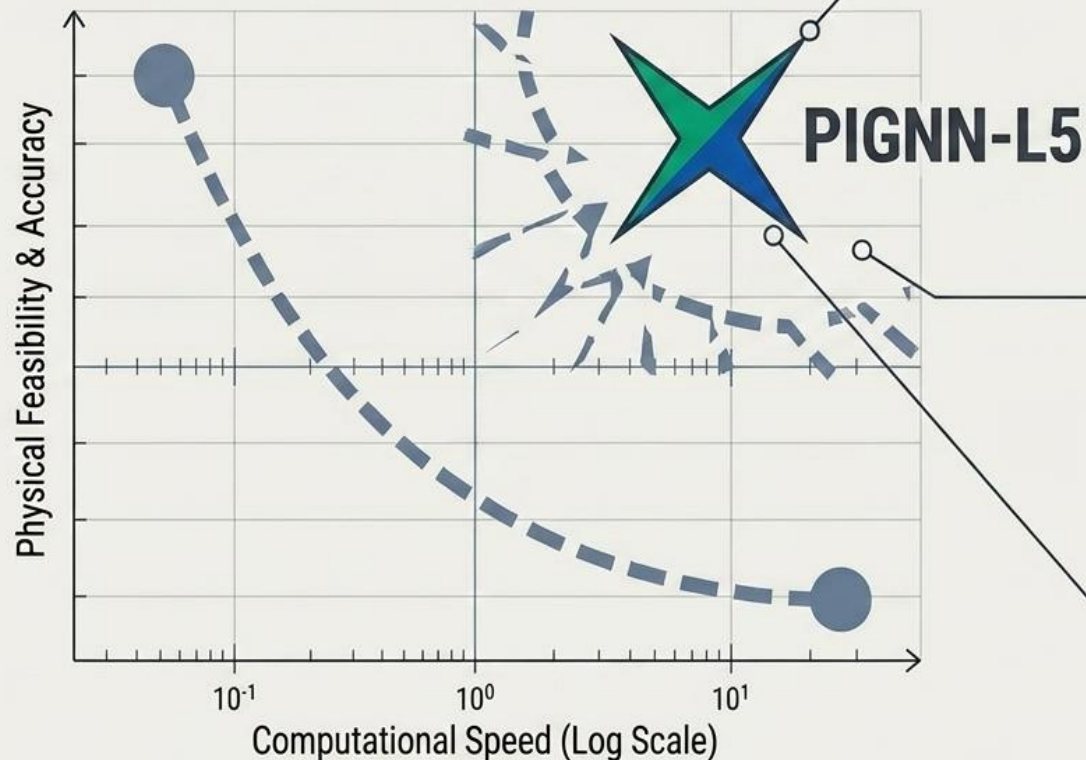
Speed vs. physical feasibility: the unreachable quadrant



Standard AI ignores physical laws; standard physics ignores computational realities.

- Newton Raphson: accurate, computationally prohibitive for scenario populations.
- Pure NNs: fast, but violate Kirchhoff balance, voltage bounds and thermal limits.
- The upper-right region, fast AND feasible and it is the operational target for edge-deployed grid analytics.

Breaking the Frontier.



Accuracy

Voltage-magnitude RMSE of 1.8×10^{-3} p.u.

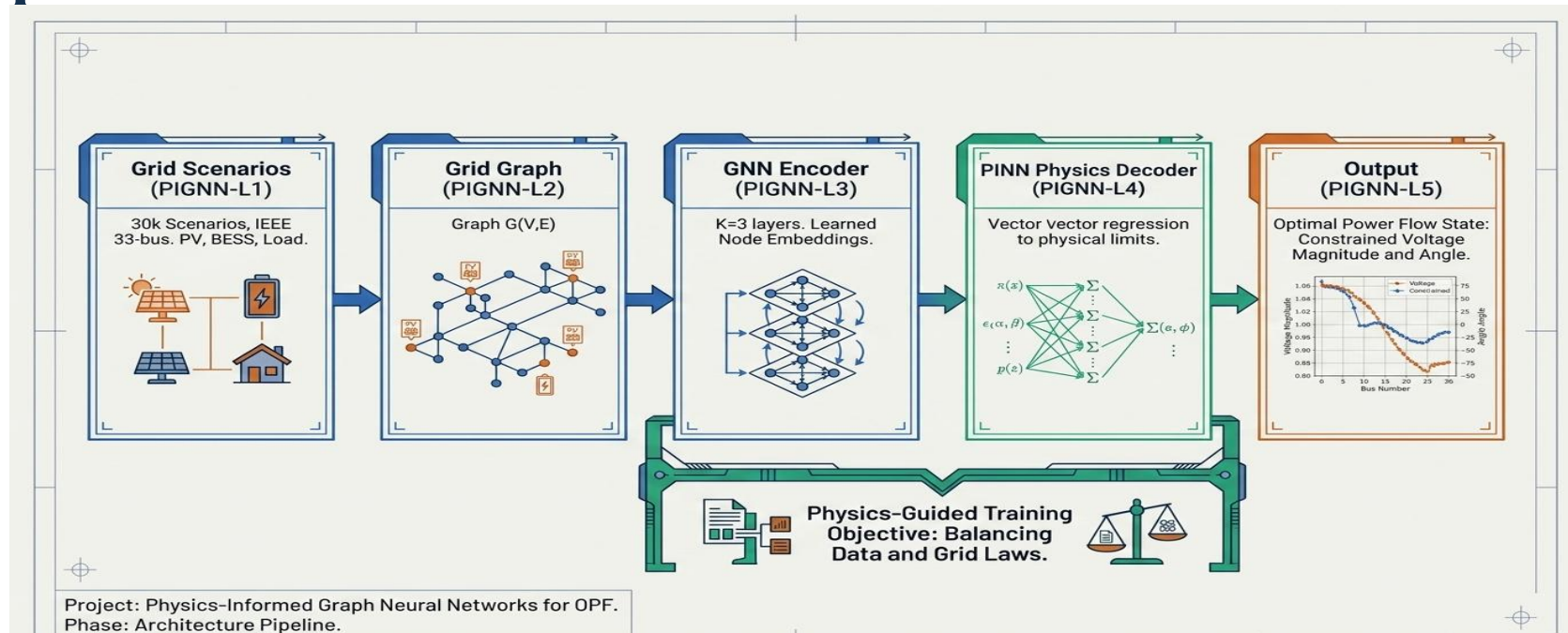
Feasibility

Suppresses active-power-balance residuals to 3.2×10^{-5} p.u.
(Voltage-bound violations <0.8%).

Speed

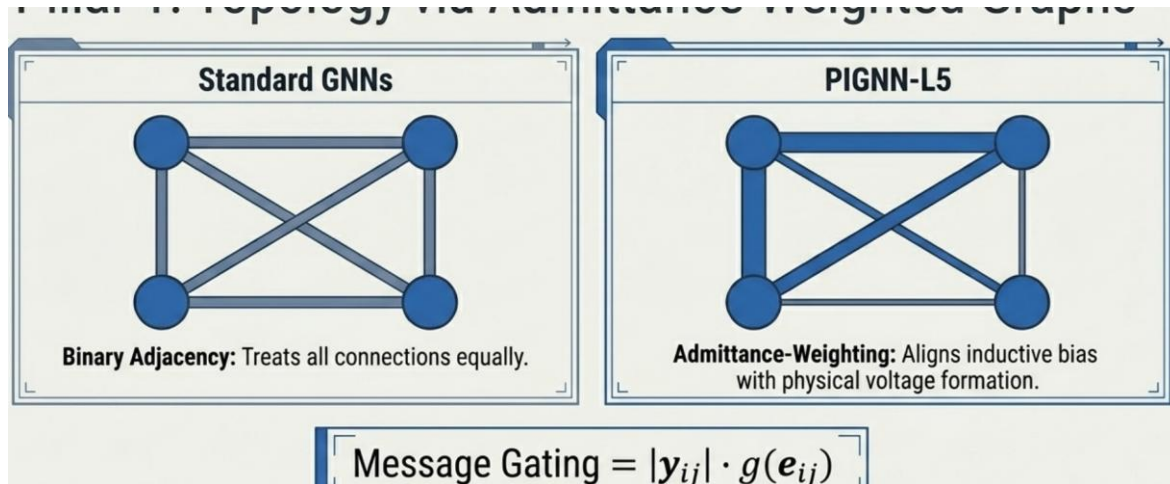
152× faster than Newton-Raphson,
executing at 0.41 ms on an
NVIDIA Jetson Orin.

PIGNN five-stage pipeline: L5 is the voltage-state prediction core



L1 scenario ingestion $\rightarrow$ L2 graph construction $\rightarrow$ L3 admittance-weighted GNN encoder ($K = 3$) $\rightarrow$ L4 physics-constrained decoder $\rightarrow$ L5 constrained voltage state $\{|V|, \delta\}$. Higher-level application layers (dispatch, anomaly detection, agentic control) are deferred to future work.

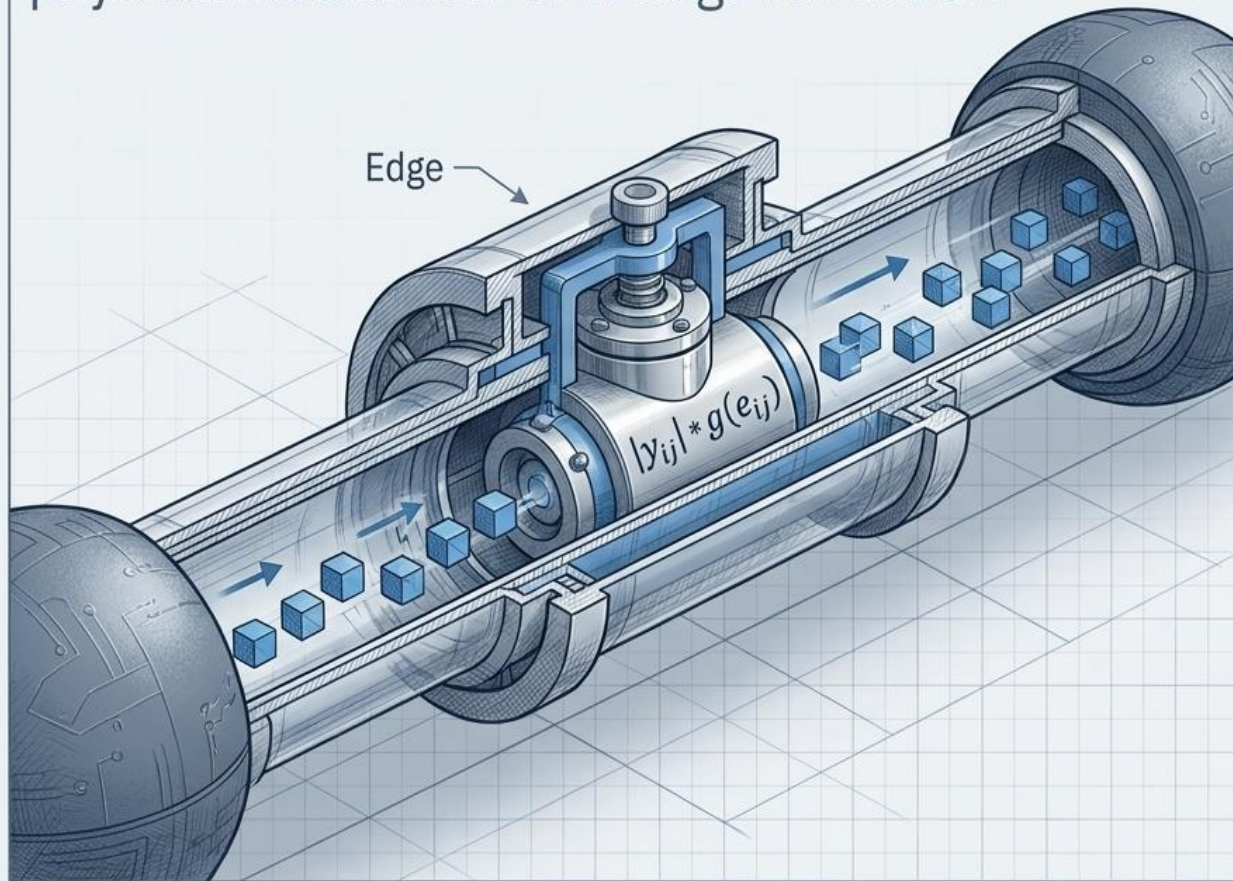
Topology via admittance-weighted message gating



- Voltage drop along a branch scales with admittance, so $|y_{ij}|$ is fixed as the dominant physical term, and only a residual gate $g(e_{ij})$ is learned on top.
- Unlike binary-adjacency or fully learned edge weights, the inductive bias is aligned with the physical mechanism of voltage formation.
- $K = 3$ message-passing rounds, hidden dim $d = 64$; residual skips ($k \geq 2$) and layer norm mitigate over-smoothing.

$$\mathbf{m}^{(k)}_i = \sum_j |y_{ij}| \cdot g(e_{ij}) \odot \mathbf{W}^{(k)} \mathbf{h}^{(k-1)}_j \quad \text{gate learned from edge features } [r_{ij}, x_{ij}, |y_{ij}|]$$

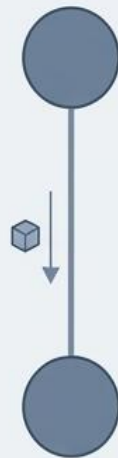
Pillar 1: Aligning inductive bias with the physical mechanism of voltage formation.



Architecture Comparison

A

Standard GNN
(Binary 1/0)



Standard GNN
(Binary 1/0)

B

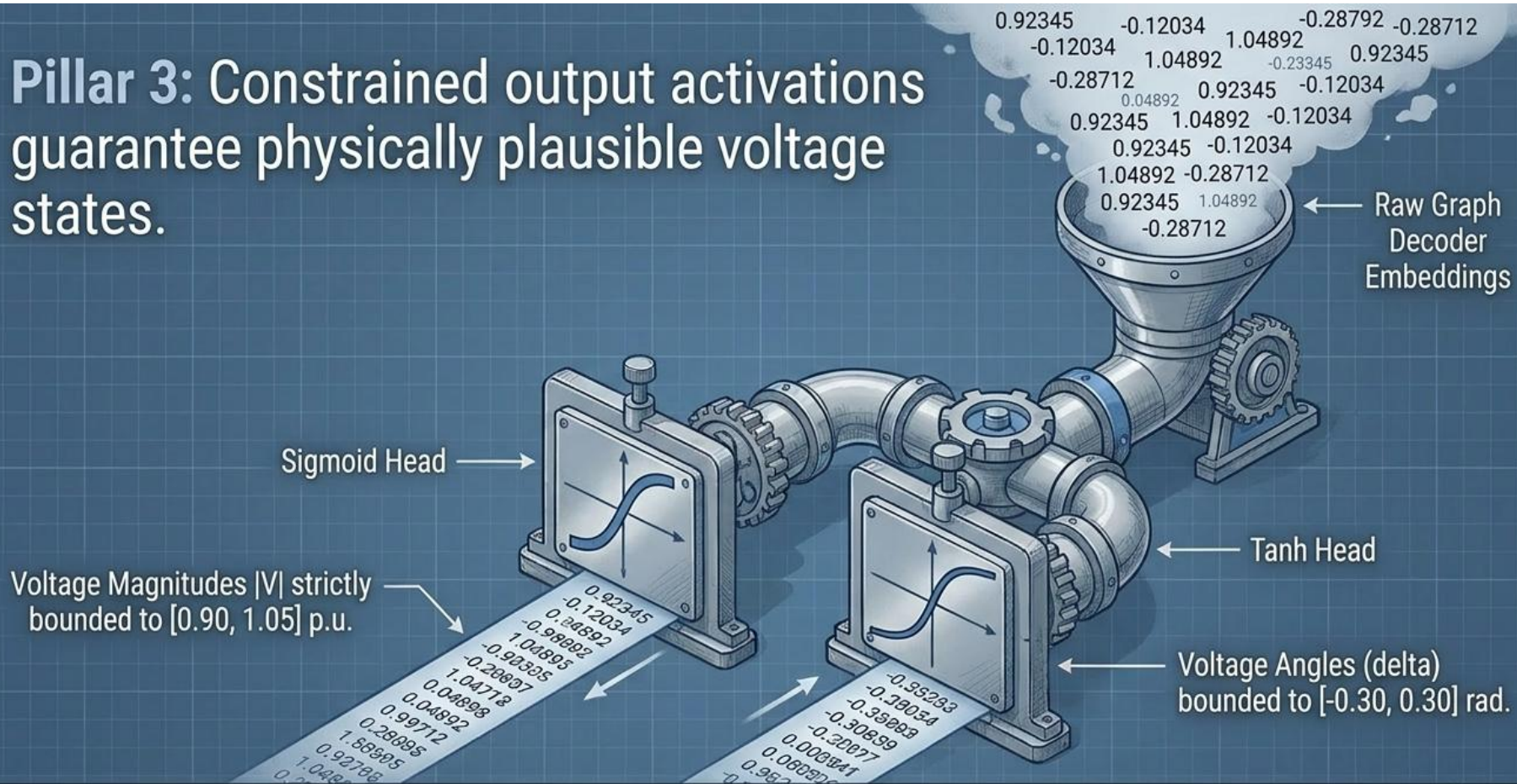
PIGNN-L5
(Admittance $|y_{ij}|$ weighted)



PIGNN-L5
(Admittance $|y_{ij}|$ weighted)

Voltage drop along a distribution branch scales directly with branch admittance. PIGNN-L5 embeds this by fixing the dominant physical term $|y_{ij}|$ and learning only a residual gate $g(e_{ij})$ on top of it.

Pillar 3: Constrained output activations guarantee physically plausible voltage states.



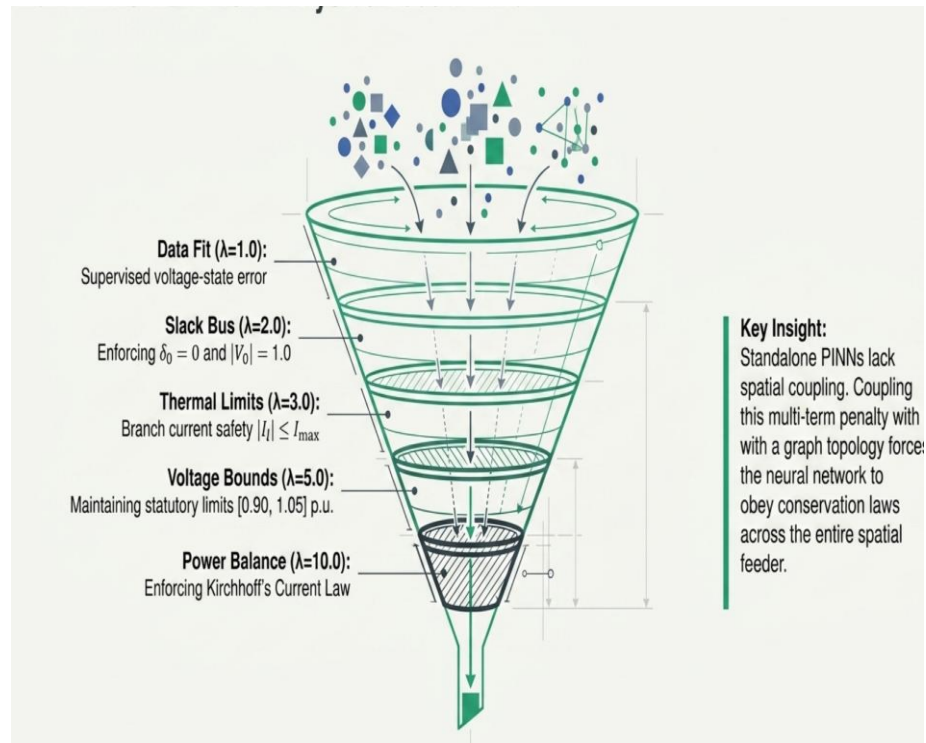
Bounded activations prevent the decoder from emitting out-of-range values during early training, improving numerical stability and dropping voltage bound violations to just 0.8%.

Five-term physics-informed training objective

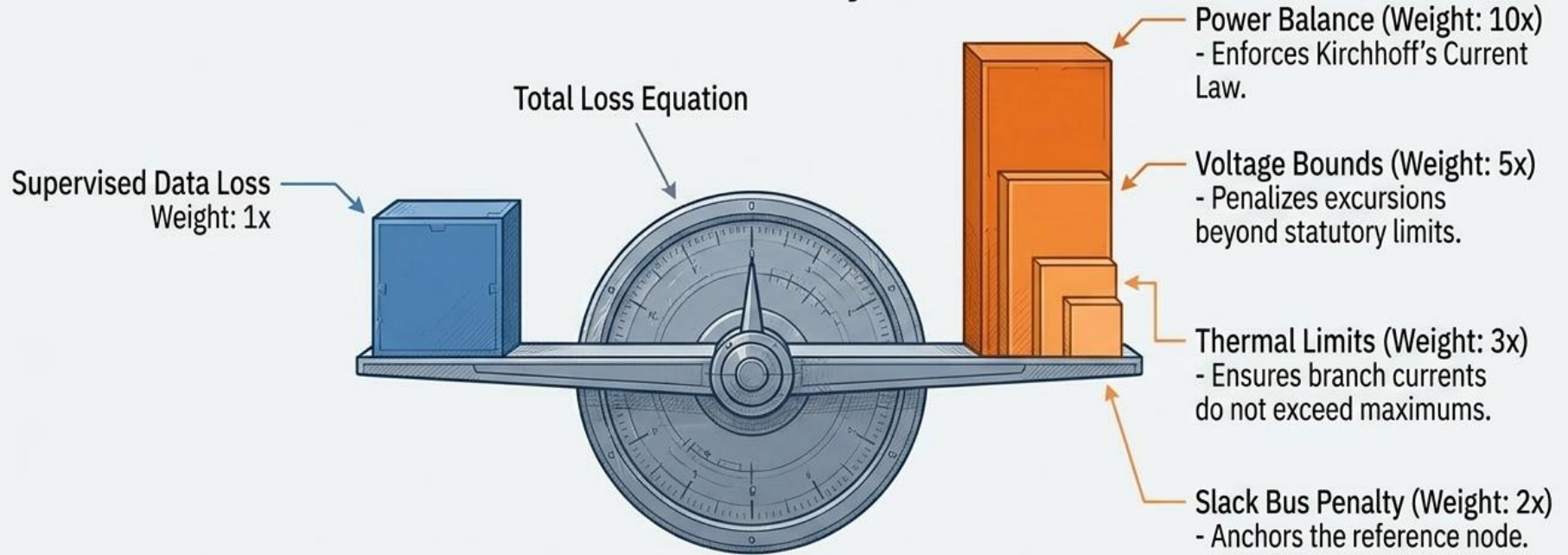
$$\mathcal{L} = \lambda_d \cdot \mathcal{L}_{\text{data}} + \lambda_{PB} \cdot \mathcal{L}_{PB} + \lambda_{VB} \cdot \mathcal{L}_{VB} + \lambda_{SL} \cdot \mathcal{L}_{SL} + \lambda_{TH} \cdot \mathcal{L}_{TH}$$

Term	λ	Enforces
Data fit	1.0	Supervised $ V $, δ error
Power balance	10.0	Kirchhoff's current law (P, Q residuals)
Voltage bounds	5.0	Statutory band [0.90, 1.05] p.u.
Thermal limits	3.0	Branch currents $ I \leq I_{\max}$
Slack bus	2.0	$\delta_0 = 0$, $ V_0 = 1.0$ p.u.

- Weights selected by coarse grid search on the validation set; λ_{PB} is dominant because power balance is the primary physical-consistency indicator.
- BESS SOC constraint deliberately excluded: snapshot-based data carries no temporal coupling: a difference-equation penalty would yield uninformative gradients.

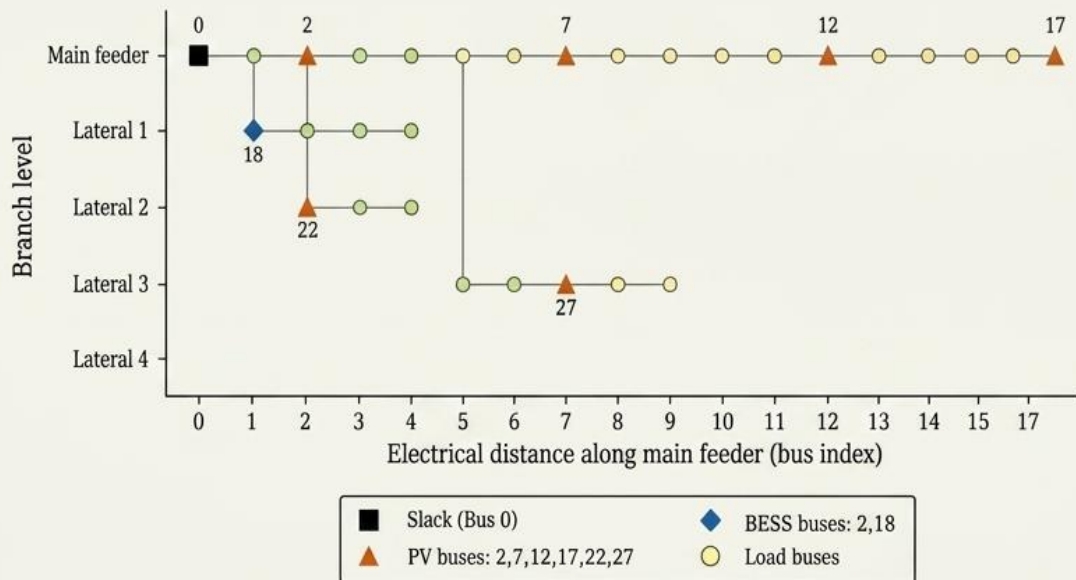


Pillar 2: A 5-term objective function enforces strict conservation-law consistency.



Power balance receives the highest weight penalty, reflecting its role as the dominant indicator of physical consistency.

The Testbed: A modified IEEE 33-bus radial active distribution feeder.



SYSTEM SCALE

33 Buses

32 Branches

6 PV Units

2 BESS Units

DATA VOLUME

30,000 Stochastic 24-hr Scenarios
(15-min resolution)

2.88 Million Operating Points

EXPERIMENTAL RIGOR

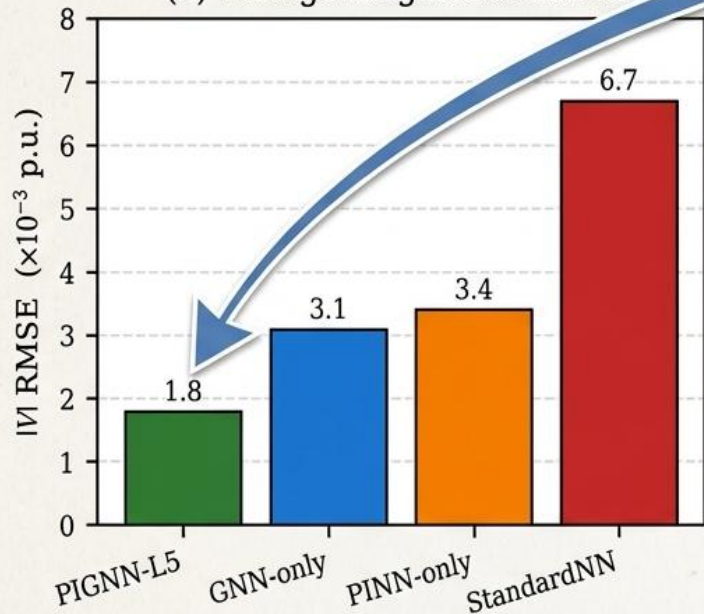
Strict 70/15/15 scenario-level split
(Train/Val/Test)

Zero temporal data leakage.

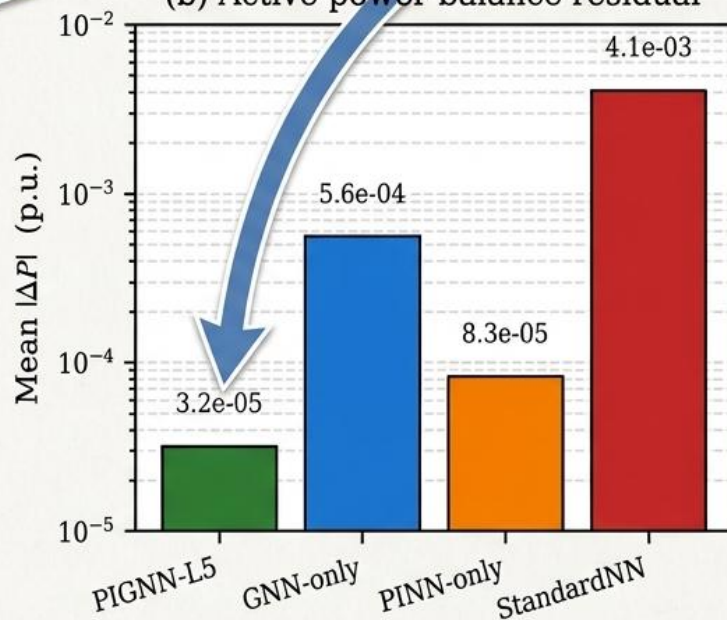
The Synthesis: Topology learning and physics regularization are complementary, not competitive.

The Ideal Frontier:
Low Error + High
Physical Feasibility

(a) Voltage-magnitude RMSE



(b) Active-power-balance residual

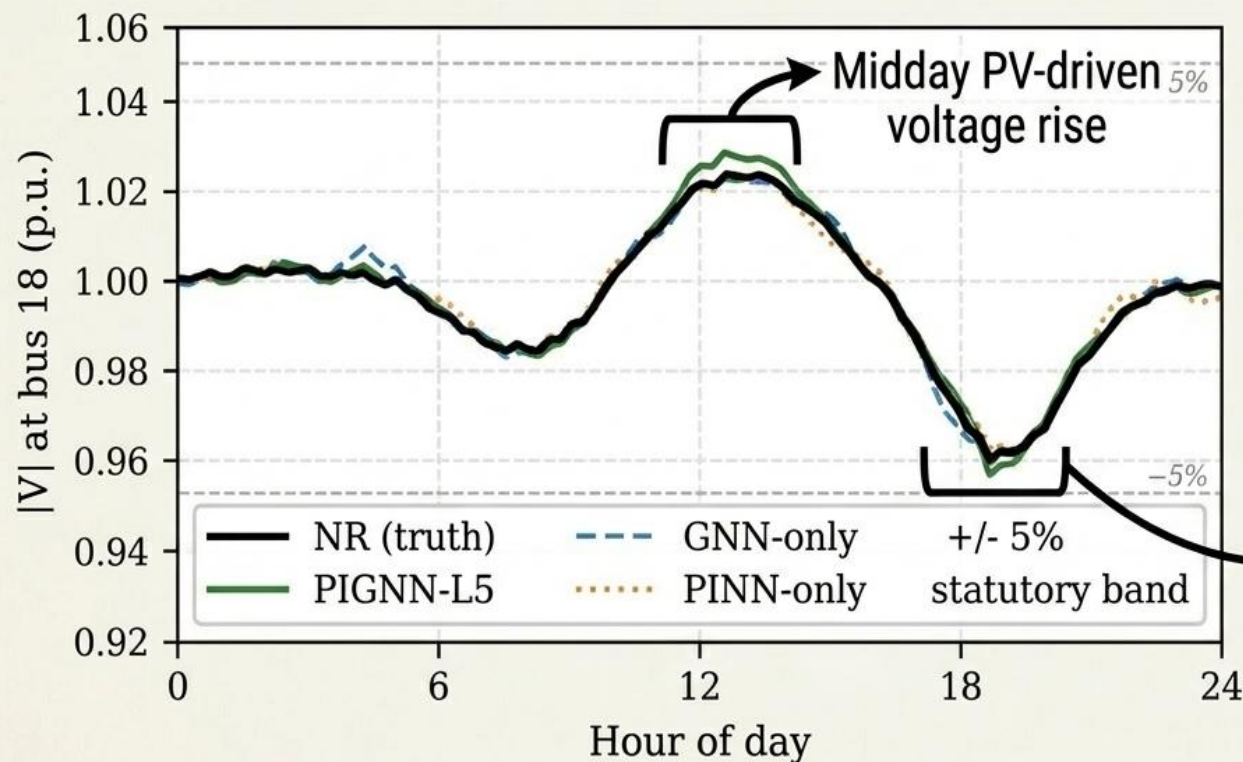


- **Accuracy:** Lowest voltage-magnitude RMSE (0.0018 p.u.)

- **Feasibility:** Suppresses active-power-balance residuals to 0.000032 p.u. (two orders of magnitude better than StandardNN).

You do not have to sacrifice spatial accuracy to achieve physical feasibility.
Embedding topology and physics synergistically solves both.

Dynamic Tracking: Flawless replication of the 24-hour voltage profile at highly sensitive buses.

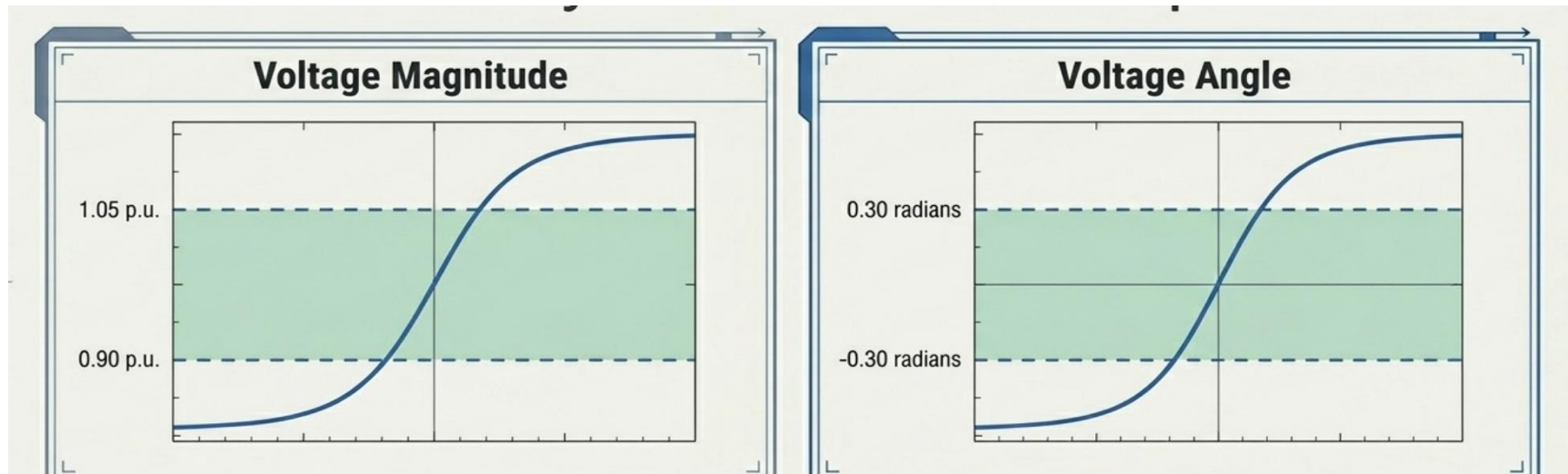


Why Bus 18?

Bus 18 sits at the furthest electrical distance from the slack bus, making it the most vulnerable to voltage excursions.

PIGNN-L5 perfectly tracks the Newton-Raphson ground truth while staying well within the $\pm 5\%$ statutory bands.

Physical plausibility via constrained output activations



sigmoid $\rightarrow |V|$

scaled to [0.90, 1.05] p.u. in-band by construction

tanh $\rightarrow \delta$

scaled to [-0.30, 0.30] rad, slack-referenced ($\delta_o = 0$)

0.8 %

voltage-bound violation rate vs 12.4 % for the unconstrained MLP

RESULTS

Held-out test split: feasibility gains with the best accuracy

Model	$ V $ RMSE (p.u.)	δ RMSE (rad)	ΔP (p.u.)	ΔQ (p.u.)	%VB	Speed-up
StandardNN (MLP)	6.7×10^{-3}	1.5×10^{-2}	4.1×10^{-3}	3.8×10^{-3}	12.4 %	185×
GNN-only	3.1×10^{-3}	7.8×10^{-3}	5.6×10^{-4}	4.9×10^{-4}	3.1 %	162×
PINN-only	3.4×10^{-3}	8.5×10^{-3}	8.3×10^{-5}	7.1×10^{-5}	2.3 %	178×
PIGNN-L5 (proposed)	1.8×10^{-3}	4.1×10^{-3}	3.2×10^{-5}	2.8×10^{-5}	0.8 %	152×
NR reference	—	—	$<10^{-6}$	$<10^{-6}$	0.0 %	1×

$\sim 100\times$

reduction in active-power-balance residual
vs the unconstrained MLP

0.8 %

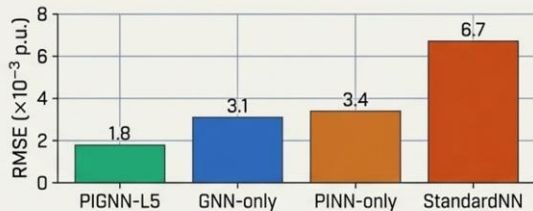
voltage-bound violations
15× lower than StandardNN (12.4 %)

1.8×10^{-3}

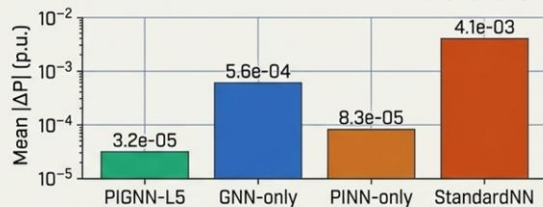
p.u. $|V|$ RMSE : lowest of all learned
models, no accuracy trade-off

24-hour tracking at bus 18: the most voltage-sensitive node

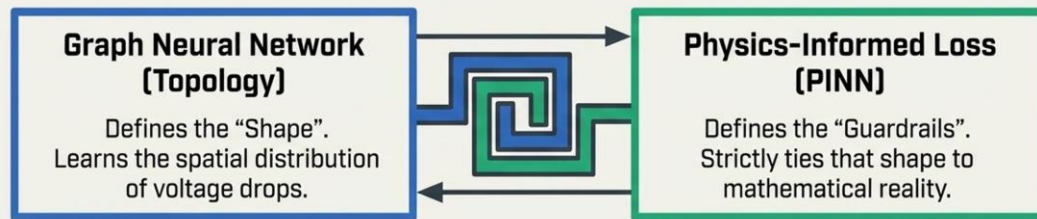
Synthesis: Shape + Guardrails



(a) Voltage-magnitude RMSE



(b) Active-power-balance residual



Core Thesis: Topology learning and physics-regularization are complementary, not substitutable. One without the other leads to mathematical chaos.

Bus 18 sits at the greatest electrical distance from the slack bus — most vulnerable to excursions.

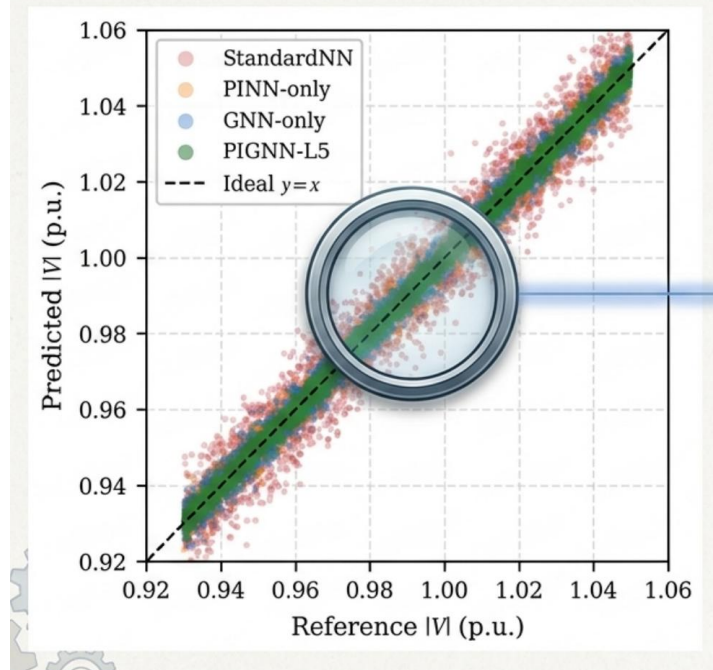
PIGNN-L5 tracks the NR reference through the full daily cycle: midday PV-driven voltage rise and the evening load-driven dip.

Predictions remain securely inside the $\pm 5\%$ statutory band on this held-out, high-PV-penetration test scenario.

Held-out aggregate: 432,000 unseen timesteps across 4,500 test scenarios.

432,000 unseen timesteps: concentration on the $y = x$ ideal

prediction.



1.8×10^{-3} p.u.

voltage-magnitude RMSE : dense green band on the diagonal

4.1×10^{-3} rad

voltage-angle RMSE, slack-referenced radian convention throughout

6.7×10^{-3} p.u.

StandardNN RMSE : the visibly wider red spread in the same plot

Only PIGNN-L5 achieves the holy trinity of speed, topology awareness, and physical feasibility.

Model	Topology Aware Graph	Multi-Constraint Physics Loss	Constrained Outputs	Sub-millisecond Speed	Voltage Feasibility Achieved
Newton-Raphson (Reference)					
StandardNN (Flat MLP)					
GNN-only (Unconstrained)					
PINN-only (No spatial coupling)					
PIGNN-L5 (Proposed)					

Standard neural models trade physics for speed. PIGNN-L5 delivers physical feasibility at an inference speed comparable to unconstrained baselines.

Dissecting the Architecture:

Every prior is mathematically indispensable.

Remove Admittance Graph
→ Spatial learning fails.
Voltage Error soars +89%

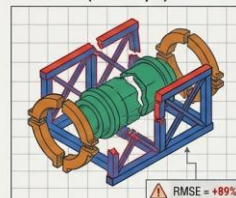
Remove Constraints
→ Outputs lose plausibility.
Voltage Bounds fail +33%

Remove Physics Loss
→ Conservation laws collapse.
Power Imbalance spikes 17x

PIGNN-L5 is not just a larger network; it is a meticulously balanced synergy of three orthogonal inductive priors.

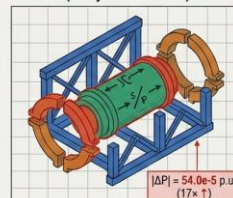
Architectural Teardown: Removing the Pillars

Remove Topology
(No Graph)



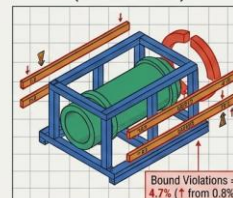
Voltage error skyrockets.
(+89% worse |V| RMSE).

Remove Physics
(Only Data Loss)



Kirchhoff compliance fails.
Active power balance residual
jumps 17x to 54.0e-5 p.u.

Remove Boundedness
(Linear Heads)



Boundary safety collapses.
Voltage bound violations
jump from 0.8% to 4.7%.

Conclusion: PIGNN-L5 is not just a larger network; it is the joint product of three orthogonal inductive priors.

Each inductive prior is indispensable for a different reason

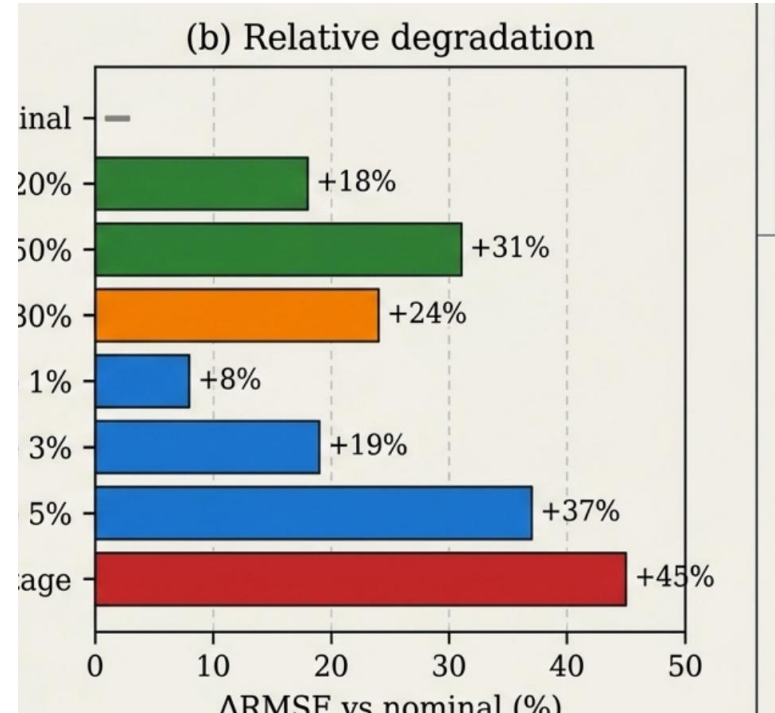
Variant	$ V $ RMSE (10^{-3} p.u.)	ΔP (10^{-5} p.u.)	%VB	Δ RMSE
Full PIGNN-L5	1.8	3.2	0.8 %	—
A1 · no graph topology	3.4	8.5	2.3 %	+89 %
A2 · no physics loss	3.1	54.0 (17×)	3.1 %	+72 %
A3 · no constrained outputs	2.4	4.1	4.7 %	+33 %
A4 · no edge electrical features	2.3	3.9	1.2 %	+28 %
A5 · K = 1 message passing	2.2	3.6	1.0 %	+22 %

Attribution is clean: topology drives accuracy (A1: +89 %), physics drives conservation-law consistency (A2: 17× worse ΔP), bounded heads drive plausibility (A3: 0.8 → 4.7 %VB).

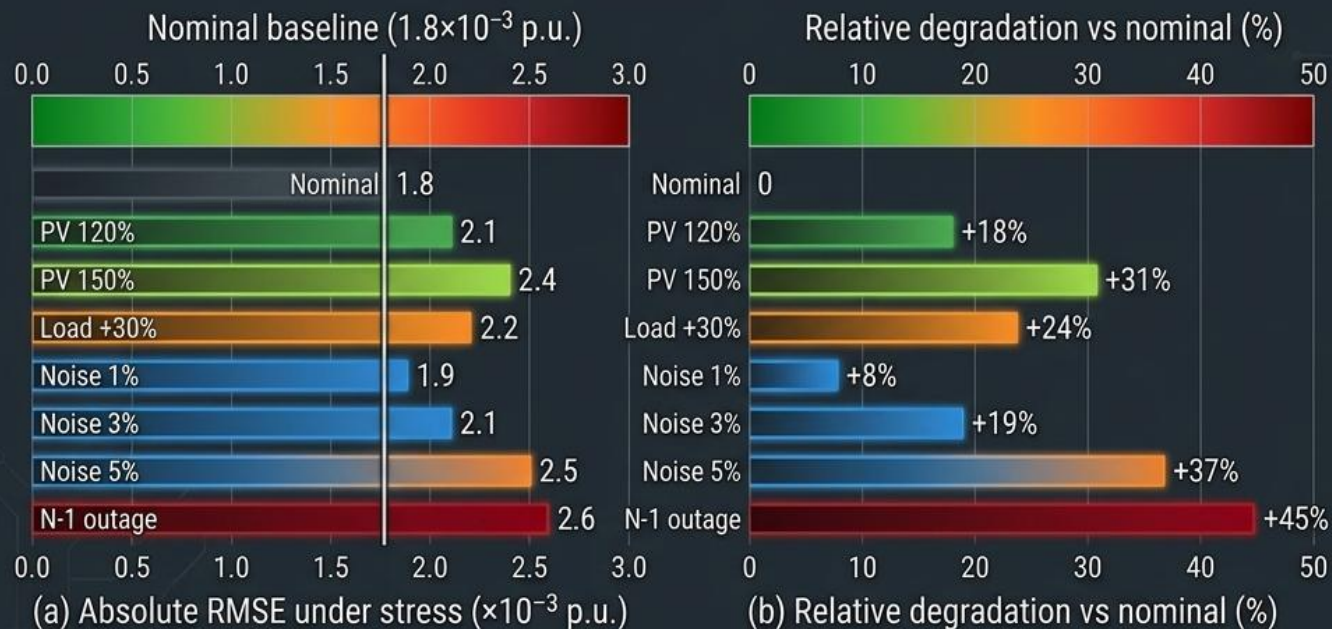
Graceful degradation : topology change is the hardest stressor

Stress condition	Δ RMSE vs nominal
R1 · PV penetration 120 % / 150 %	+18 % / +31 %
R2 · Load +30 %	+24 %
R3 · Sensor noise 1 % / 3 % / 5 %	+8 % / +19 % / +37 %
R4 · N-1 line outage (topology)	+45 %

- Noise response is smooth and monotonic, the surrogate does not collapse abruptly under input corruption.
- N-1 changes the feeder graph itself relative to fixed-topology training, cleanly isolating topology-adaptive learning as the next research frontier.



Graceful Degradation: Surviving out-of-distribution grid stress without retraining



+150% PV & +30% Load

Handled with moderate (+24–31%) degradation.

5% Additive Noise

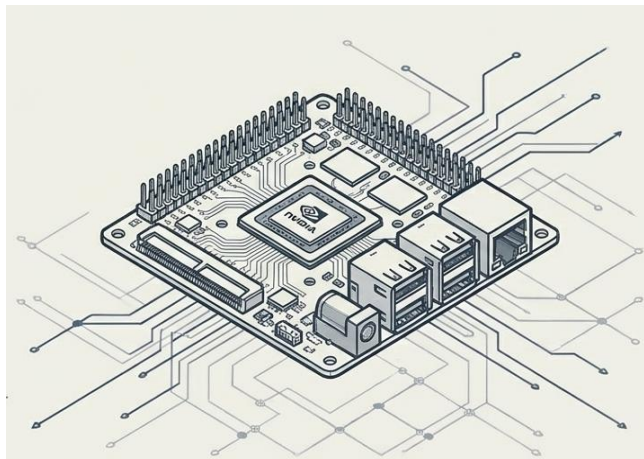
Smooth, non-collapsing performance (+37% error).

N-1 Line Outage

The ultimate stressor. Modifying the physical topology causes the highest degradation (+45%).

Modifying the graph topology (N-1) is definitively the hardest stressor, identifying topology-adaptive learning as the ultimate frontier.

Sub-millisecond inference, trained and timed on NVIDIA Jetson Orin

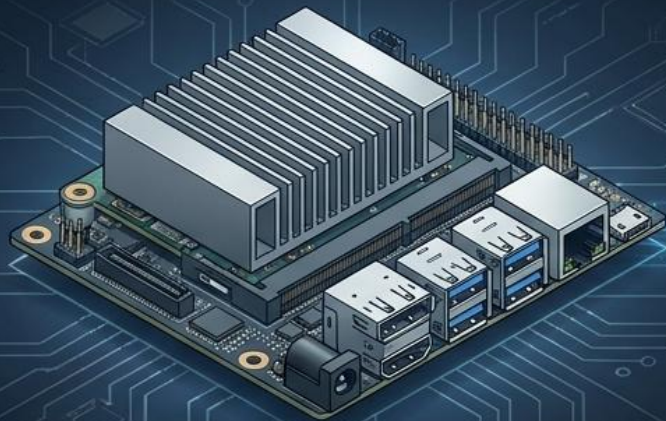


Platform	
NVIDIA Jetson Orin (Edge IoT environment)	
Parameter Count	
309,378 (Extremely lightweight)	
Inference Latency	
0.41 milliseconds per snapshot	
Acceleration	
152× faster than pandapower pure-Python Newton-Raphson reference	

Model	ms / snapshot	Speed-up
NR reference (pandapower)	62.3	1×
StandardNN	0.34	185×
GNN-only	0.38	162×
PINN-only	0.35	178×
PIGNN-L5	0.41	152×

- 309,378 trainable parameters : dominated by the flatten-to-decoder projection; lightweight enough for embedded platforms.
- The 20–25 % overhead of admittance-weighted message passing buys an order-of-magnitude reduction in feasibility violations.
- **Honest framing: 152× is vs a pure-Python NR stack : an order-of-magnitude indication, not a hardware-optimal bound. The value is feasibility at unconstrained-surrogate speed.**

Edge Computing Viability: 152x faster than traditional Newton-Raphson solvers



1



Hardware: Deployed natively on an NVIDIA Jetson Orin edge platform.

2



Speed: 0.41 milliseconds per snapshot inference time.

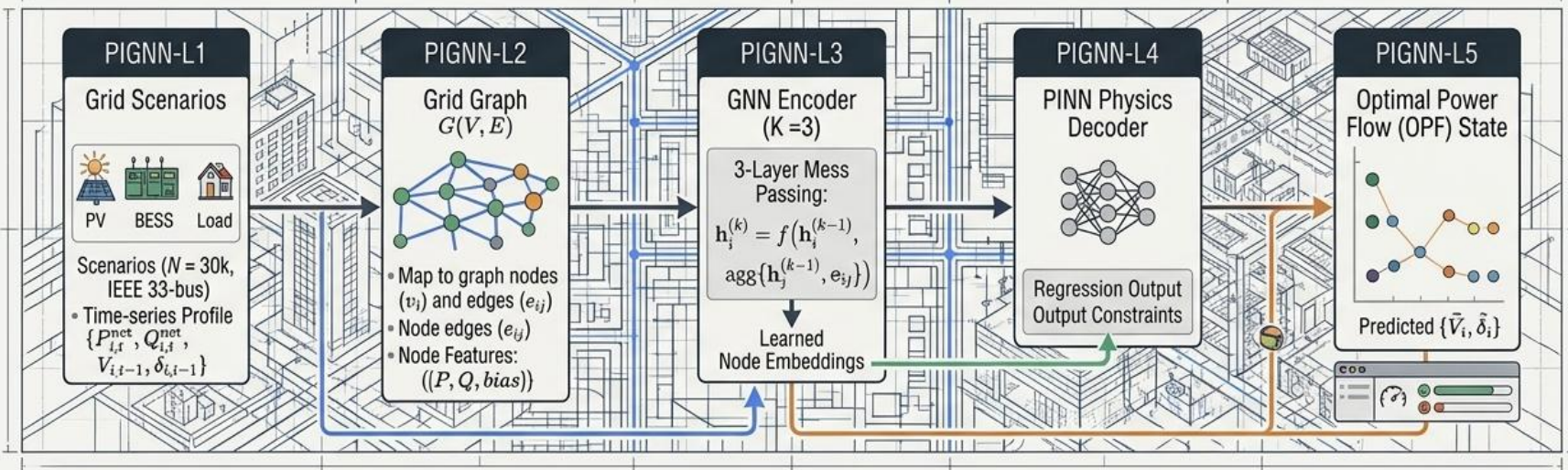
3



Efficiency: Highly compact at just 309,378 trainable parameters.

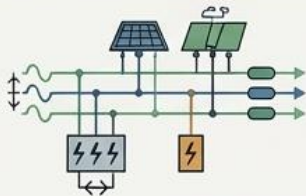
PIGNN-L5 delivers the physical feasibility of heavy Newton-Raphson simulation at the sub-millisecond speed of an unconstrained neural network—fully deployable on local edge devices.

The Blueprint for Topology-Adaptive Grids



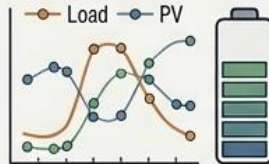
Horizon 1 | Three-Phase Reality

Extending to unbalanced feeders with mutual coupling.



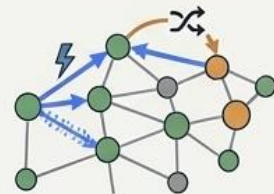
Horizon 2 | Temporal Dynamics

Multi-period forecasting integrating BESS state-of-charge.



Horizon 3 | Topology Adaptation

Online graph updating for real-time N-k contingency switching.



The core problem of distribution-grid power flow is no longer speed. With **PIGNN-L5**, it is speed bounded by physical law.

Speed bounded by physical law

Topology

Admittance-weighted gating $|y_{ij}| \cdot g(e_{ij})$ drives spatial accuracy (1.8×10^{-3} p.u.)

Physics

Five-term objective suppresses ΔP to 3.2×10^{-5} p.u. $\sim 100\times$ below the MLP baseline

Plausibility

Constrained heads hold voltage-bound violations at 0.8 % by construction

Future work

- Three-phase unbalanced feeders with explicit mutual coupling
- Multi-period formulations preserving BESS state-of-charge, ramp limits and demand response
- PIGNN-L5 as a differentiable power-flow layer inside Volt/VAR, hosting-capacity and contingency pipelines
- Topology-adaptive fine-tuning / meta-learning for N-k contingency operation, the priority identified by the +45 % N-1 result

Reproducibility

Source code, pretrained weights, scenario-generation scripts and evaluation notebooks to be released for native Jetson Orin replication.